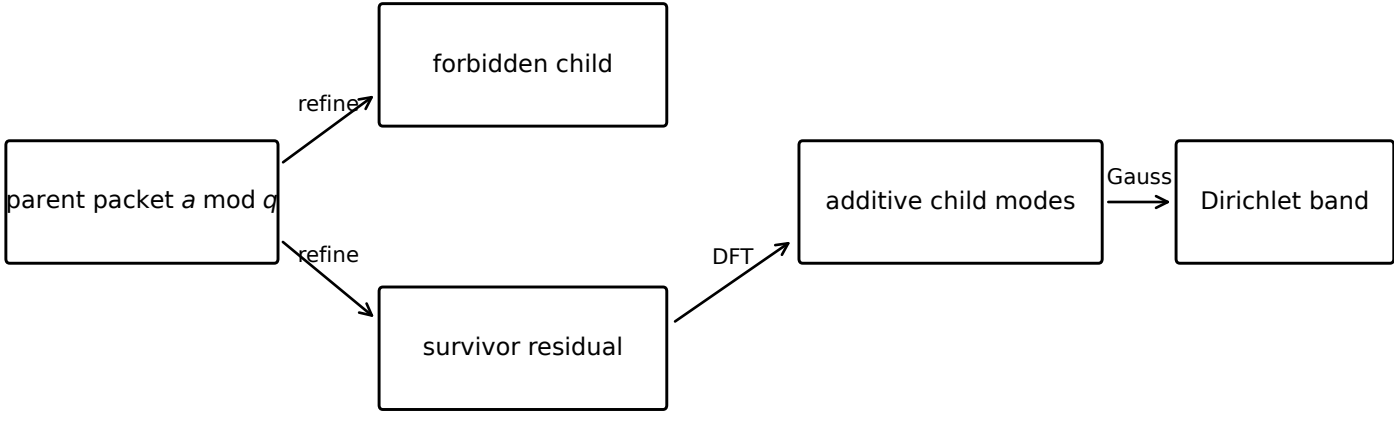Arithmetic realization: sieve geometry and character energy are changes of resolution